

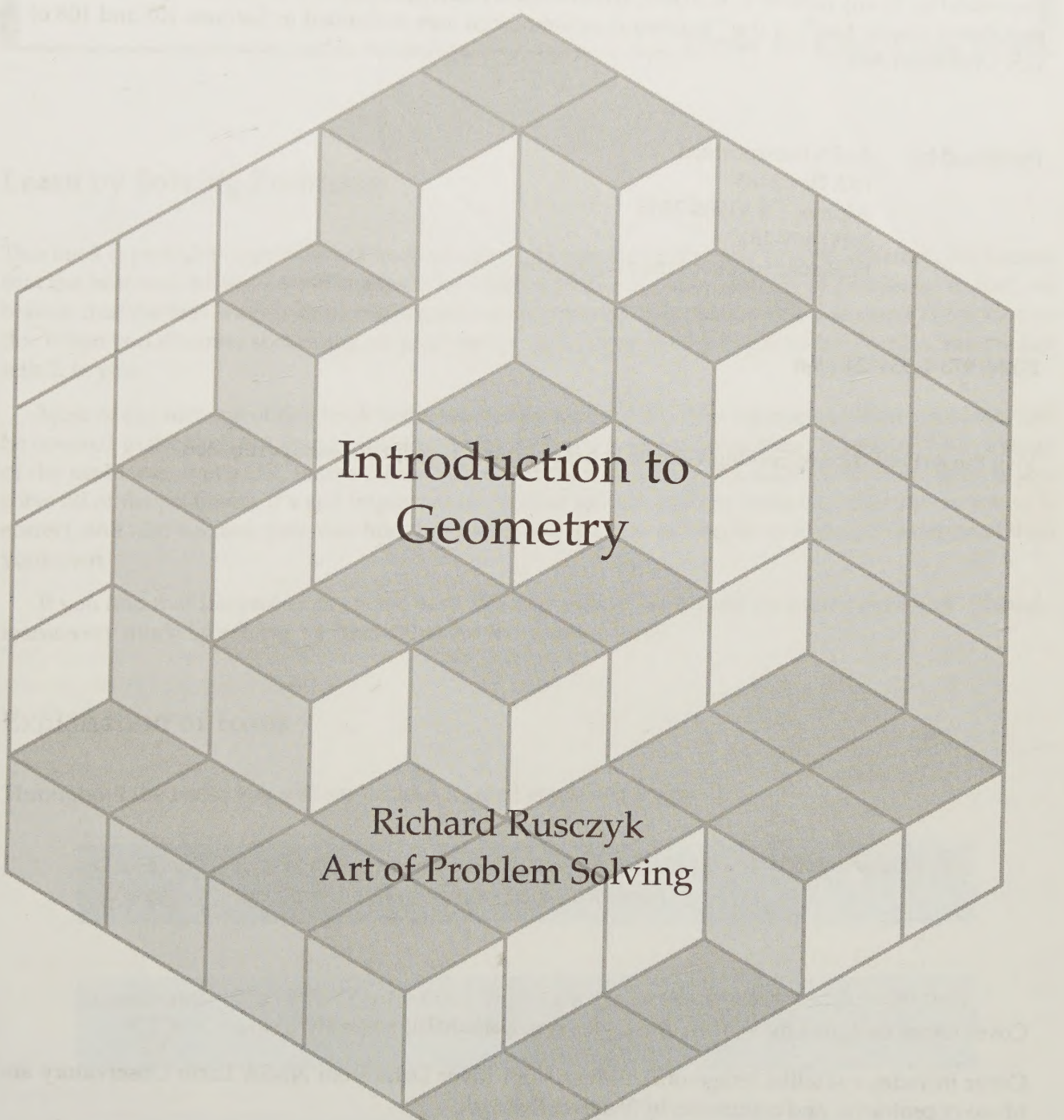
the Art of Problem Solving

Introduction to Geometry

Richard Rusczyk



2nd Edition



Introduction to Geometry

Richard Rusczyk
Art of Problem Solving

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How to Use This Book

Learn by Solving Problems

This book is probably very different from most of the math books that you have read before. We believe that the best way to learn mathematics is by solving problems. Lots and lots of problems. In fact, we believe that the best way to learn mathematics is to try to solve problems that you don't know how to do. When you discover something on your own, you'll understand it much better than if someone just tells it to you.

Most of the sections of this book begin with several problems. The solutions to these problems will be covered in the text, but try to solve the problems *before* reading the section. If you can't solve some of the problems, that's OK, because they will all be fully solved as you read the section. Even if you solve all of the problems, it's still important to read the section, both to make sure that your solution is correct, and also because you may find that the book's solution is simpler or easier to understand than your own.

If you find that the problems are too easy, this means that you should try harder problems. Nobody learns very much by solving problems that are too easy for them.

Explanation of Icons

Throughout the book, you will see various shaded boxes and icons.

Concept: This will be a general problem-solving technique or strategy. These are the "keys" to becoming a better problem solver!



Important: This will be something important that you should learn. It might be a formula, a solution technique, or a caution.



WARNING!! Beware if you see this box! This will point out a common mistake or pitfall.



Game: Remember, math is fun! This box will contain a game to think about.



Sidenote: This box will contain material which, although interesting, is not part of the main material of the text. It's OK to skip over these boxes, but if you read them, you might learn something interesting!



Bogus Solution: Just like the impossible cube shown to the left, there's something wrong with any "solution" that appears in this box.



Exercises, Review Problems, and Challenge Problems

Most sections end with several **Exercises**. These will test your understanding of the material that was covered in the section that you just finished. You should try to solve *all* of the exercises. Exercises marked with a ★ are more difficult.

Most chapters have several **Review Problems**. These are problems which test your basic understanding of the material covered in the chapter. Your goal should be to solve most or all of the Review Problems for every chapter – if you're unable to do this, it means that you haven't yet mastered the material, and you should probably go back and read the chapter again.

All of the chapters end with **Challenge Problems**. These problems are generally more difficult than the other problems in the book, and will really test your mastery of the material. Some of them are very, very hard – the hardest ones are marked with a ★. Don't necessarily expect to be able to solve all of the Challenge Problems on your first try – these are difficult problems even for experienced problem solvers. If you are able to solve a large number of Challenge Problems, then congratulations, you are on your way to becoming an expert problem solver!

Hints

Many problems come with one or more hints. You can look up the hints in the Hints section in the back of the book. The hints are numbered in random order, so that when you're looking up a hint to a problem you don't accidentally glance at the hint to the next problem at the same time.

It is very important that you first try to solve the problem without resorting to the hints. Only after you've seriously thought about a problem and are stuck should you seek a hint. Also, for problems which have multiple hints, use the hints one at a time; don't go to the second hint until you've thought about the first one.

Solutions

The solutions to all of the Exercises, Review Problems, and Challenge Problems are in the separate solution book. If you are using this textbook in a regular school class, then your teacher may decide not to make this solution book available to you, and instead present the solutions him/herself. However, if you are using this book on your own to learn independently, then you probably have a copy of the solution book, in which case there are some very important things to keep in mind:

1. Make sure that you make a serious attempt at the problem before looking at the solution. Don't use the solution book as a crutch to avoid really thinking about a problem first. You should think *hard* about a problem before deciding to give up and look at the solution.
2. After you solve a problem, it's usually a good idea to read the solution, even if you think you know how to solve the problem. The solution that's in the solution book might show you a quicker or more concise way to solve the problem, or it might have a completely different solution method that you might not have thought of.
3. If you have to look at the solution in order to solve a problem, make sure that you make a note of that problem. Come back to it in a week or two to make sure that you are able to solve it on your own, without resorting to the solution.

Resources

Here are some other good resources for you to further pursue your study of mathematics:

- *The Art of Problem Solving* books, by Sandor Lehoczky and Richard Rusczyk. Whereas the book that you're reading right now will go into great detail of one specific subject area – geometry – *the Art of Problem Solving* books cover a wide range of problem solving topics across many different areas of mathematics.
- The www.artofproblemsolving.com website. The authors of this book are also the webmasters of the Art of Problem Solving website, which contains many resources for students:
 - a discussion forum
 - online classes
 - resource lists of books, contests, and other websites
 - a $\text{\LaTeX}$ tutorial
 - and much more!
- You can hone your problem solving skills (and perhaps win prizes!) by participating in various math contests. For middle school students in the United States, the major contests are MATHCOUNTS, MOEMS, and the AMC 8. For U.S. high school students, some of the best-known contests are the AMC/AIME/USAMO series of contests (which choose the U.S. team for the International

Mathematics Olympiad), the American Regions Math League (ARML), the Mandelbrot Competition, the Harvard-MIT Mathematics Tournament, and the USA Mathematical Talent Search. More details about these contests are on page vii, and links to these and many other contests are available on the Art of Problem Solving website.

A Note to Teachers

We believe that students learn best when they are challenged with hard problems that at first they may not know how to do. This is the motivating philosophy behind this book.

Rather than first introducing new material and then giving students exercises, we present problems at the start of each section that students should try to solve *before* the new material is presented. The goal is to get students to discover the new material on their own. Often, complicated problems are broken into smaller parts, so that students can discover new techniques one piece at a time. Then the new material is formally presented in the text, and full solutions to each problem are explained, along with problem-solving strategies.

We hope that teachers will find that their stronger students will discover most of the material in this book on their own by working through the problems. Other students may learn better from a more traditional approach of first seeing the new material, then working the problems. Teachers have the flexibility to use either approach when teaching from this book.

The book is linear in coverage. Generally, students and teachers should progress straight through the book in order, without skipping chapters. Sections denoted with a ★ contain supplementary material that may be safely skipped. In general, chapters are not equal in length, so different chapters may take different amounts of classroom time.

Extra! Occasionally, you'll see a box like this at the bottom of a page. This is an "Extra!" and might be a quote, some biographical or historical background, or perhaps an interesting idea to think about.

Acknowledgements

Contests

We would like to thank the following contests for allowing us to use a selection of their problems in this book:

- The **American Mathematics Competitions**, a series of contests for U.S. middle and high school students. The **AMC 8**, **AMC 10**, and **AMC 12** contests are multiple-choice tests which are taken by over 400,000 students every year. Top scorers on the AMC 10 and AMC 12 are invited to take the **American Invitational Mathematics Examination (AIME)**, which is a more difficult, short-answer contest. Approximately 10,000 students every year participate in the AIME. Then, based on the results of the AMC and AIME contests, about 250 students are invited to participate in the **USA Mathematical Olympiad (USAMO)**, a 2-day, 9-hour examination in which each student must show all of his or her work. Results from the USAMO are used to invite a number of students to the Math Olympiad Summer Program, at which the U.S. team for the International Mathematical Olympiad (IMO) is chosen. More information about the AMC contests can be found on the AMC website at <http://www.unl.edu/amc>.
- **MATHCOUNTS®**, the premier contest for U.S. middle school students. MATHCOUNTS is a national enrichment, coaching, and competition program that promotes middle school mathematics achievement through grassroots involvement in every U.S. state and territory. President George W. Bush and former Presidents Clinton, Bush and Reagan have all recognized MATHCOUNTS in White House ceremonies. The MATHCOUNTS program has also received two White House citations as an outstanding private sector initiative. Particularly exciting for MATHCOUNTS Mathletes® were the hour-long ESPN programs on each of the past three National Competitions. More information is available at www.mathcounts.org.
- The **Mandelbrot Competition**, which was founded in 1990 by Sandor Lehoczky, Richard Rusczyk, and Sam Vandervelde. The aim of the Mandelbrot Competition is to provide a challenging, engaging mathematical experience which is both competitive and educational. Students compete both as individuals and in teams. The Mandelbrot Competition is offered at the national level for more advanced students and the regional level for less experienced problem solvers. More information can be found at www.mandelbrot.org.
- The **Harvard-MIT Mathematics Tournament (HMMT)**, which is an annual math tournament for high school students, held at MIT and at Harvard in alternating years. It is run exclusively by MIT and Harvard students, most of whom themselves participated in math contests in high school. More information is available at web.mit.edu/hmmt/.

- The **USA Mathematical Talent Search (USAMTS)**, which was founded in 1989 by Professor George Berzsenyi. The USAMTS is a free mathematics competition open to all United States middle and high school students. As opposed to most mathematics competitions, the USAMTS allows students a full month to work out their solutions. Carefully written justifications are required for each problem. More information is available at www.usamts.org.
- The **American Regions Math League (ARML)**, which was founded in 1976. The annual ARML competition brings together nearly 2,000 of the nation's finest students. They meet, compete against, and socialize with one another, forming friendships and sharpening their mathematical skills. The contest is written for high school students, although some exceptional junior high students attend each year. The competition consists of several events, which include a team round, a power question (in which a team solves proof-oriented questions), an individual round, and two relay rounds. More information is available at www.arml.com.

How We Wrote This Book

This book was written using the $\text{\LaTeX}$ document processing system. Specifically, this book was prepared using the MiKTeX installation of pdf $\text{\LaTeX}$ on a PC running Microsoft Windows XP. We must thank the authors of the various $\text{\LaTeX}$ packages that we used while preparing this book, and also the brilliant authors of *The $\text{\LaTeX}$ Companion* for writing a reference book that is not only thorough but also very readable. The diagrams were prepared using METAPOST, a powerful graphics language which is based on Knuth's METAFONT.

About Us

This book is a collaborative effort of the staff of the Art of Problem Solving. Richard Rusczyk was the lead author for this book, and wrote most of the text. Some of the Sidebar and Extra sections were prepared by Ashley Reiter Ahlin, Vanessa Rusczyk, and Naoki Sato. The solutions were written by Ruozhou Jia, Brian Rice, Richard Rusczyk, and Naoki Sato. Extensive proofreading of the manuscript was done by Mathew Crawford, Lisa Davis, Amanda Jones, David Patrick, Tim Lambert, Naoki Sato, and Jake Wildstrom. Vanessa Rusczyk designed the cover and also contributed greatly to the interior design of the book. David Patrick, Naoki Sato, Ravi Boppana, Meena Boppana, Valentin Vornicu, Greg Brockman, Larry Evans, and Joseph Laurendi contributed problems and proofreading to this second edition.

The author would also like to thank Josh Zucker, whose comments about how he learned mathematics inspired the questions-before-the-lessons approach of the text.

Websites & Errata

We used several websites as source material for the text. Links to these sites, as well as an errata list for the text and solutions, are provided at

<http://www.artofproblemsolving.com/BookLinks/IntroGeometry/links.php>

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For **Professor Harold Reiter**, who brought me back to education, and for **Vanessa Rusczyk**, whose confidence in me and love of the desert has kept me here ever since.